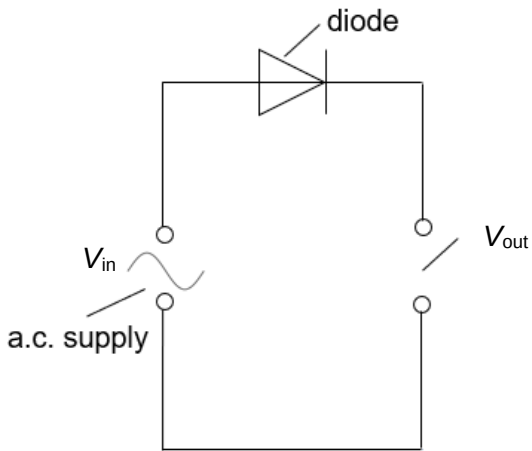


Q7	
(a)(i)	$B_C = 7.5 \times 10^{-3} \text{ T}$
(a)(ii)	$B = \frac{\mu_0 NI}{2r}$ $7.5 \times 10^{-3} = \frac{\mu_0 (300)(2.0)}{2(0.050)}$ $\mu_0 = 1.25 \times 10^{-6} \text{ H m}^{-1}$
(b)(i)	The <u>magnitude of the induced e.m.f.</u> in a conductor is <u>directly proportional</u> to the <u>rate of change of magnetic flux linkage</u> experienced by the conductor.
(b)(ii)	0.024 – 0.036 m
(b)(iii)	The gradient of the graph is the maximum. As coil Q is moved at a steady speed, the rate of change of magnetic flux linkage is the greatest. Hence induced e.m.f. is maximum.
(b)(iv)	When coil Q is moved to the right, the magnetic flux will decrease. By Lenz's law, an induced current caused by an induced e.m.f. in coil Q will flow in the same direction as the current in coil P to create a magnetic field to the left.
(b)(v)1	$B \text{ at } 0.040 \text{ m} = 3.6 \times 10^{-3} \text{ T}$ $\Delta B = 3.6 \times 10^{-3} - 7.5 \times 10^{-3}$ $= -3.9 \times 10^{-3} \text{ T}$
(b)(v)2	$\mathcal{E}_{\text{ave}} = -\frac{d\phi}{dt}$ $= -\frac{dNBA \cos \theta}{dt}$ $= -NA \frac{dB}{dt}$ $= -\frac{(5000)(1.5 \times 10^{-4})(-3.9 \times 10^{-3})}{0.25}$ $= 1.17 \times 10^{-2} \text{ V}$
(c)(i)	Value of an alternating current that is equal to the <u>steady direct current</u> that would <u>dissipate heat at the same average rate in a given resistor</u> .
(c)(ii)1	

	$P_0 = 2P_{\text{ave}}$ $= 2(20)$ $= 40 \text{ W}$
	Correct shape for 2 complete cycles
(c)(ii)2	$\frac{V_{0S}}{V_{0P}} = \frac{N_S}{N_P}$ $\frac{V_{0S}}{240} = \frac{1}{50}$ $V_{0S} = 4.8 \text{ V}$
	$V_{\text{r.m.s. S}} = \frac{4.8}{\sqrt{2}}$ $= 3.39 \text{ V (3 s.f.)}$
(c)(iii)	
	Labelled circuit diagram with correct symbols showing V_{in} , V_{out} , a.c. supply and diode.
(c)(iv)	$V_{\text{r.m.s. half-wave}} = \frac{4.8}{2}$ $= 2.4 \text{ V}$